FinRAG: RAG-Powered Financial Q&A with Hallucination Detection

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Repository: https://github.com/pendyal1/Financial_QA_ChatBot

Abstract

SEC filings are among the most authoritative sources of public financial information, but their length and density make them difficult to search manually. General-purpose language models can answer financial questions fluently but often produce responses that are not grounded in actual documents, which is a particularly dangerous failure mode in finance. This paper presents FinRAG, a retrieval-augmented generation system that answers natural language questions about public-company SEC filings by pulling filing content live from the SEC EDGAR database, retrieving the most relevant passages using hybrid search and cross-encoder reranking, and generating answers grounded in cited evidence. A hallucination detection module evaluates each answer against retrieved sources and assigns a confidence score and risk label. We also fine-tune Qwen2.5-7B-Instruct using QLoRA on a curated mix of FinQA, ConvFinQA, and TAT-QA data, drawing on strategies from domain-adapted financial LLMs such as DragonLLM's Qwen-Open-Finance-R-8B. Case studies demonstrate reliable hallucination control across numeric, risk, and competitive-landscape queries. An ablation study isolates the contribution of each pipeline stage. Results confirm that retrieval grounding reduces hallucination risk substantially, but also reveal that citation support alone does not guarantee answer quality. Synthesis, ordering, and formatting remain the next critical challenges.

1. Introduction

Financial filings like 10-K and 10-Q reports are critical documents in public markets. They contain a company's risk disclosures, revenue figures, competitive positioning, and management commentary, all written to satisfy legal disclosure requirements. In theory, anyone can access them. In practice, a single 10-K can run over 100 pages, and the relevant section for any particular question may be buried deep within. Investors, analysts, students, and researchers all spend significant time doing this search manually.

Large language models offer a natural solution: ask a question, get an answer. The problem is that general-purpose models frequently hallucinate in high-stakes domains. They generate confident-sounding responses that are not grounded in any real document. In finance, that is dangerous. A model that states the wrong revenue figure, invents a risk factor, or misattributes a quote could mislead decisions with real financial consequences.

FinRAG addresses this by combining retrieval-augmented generation with hallucination detection. Instead of relying on a model's parametric memory, the system pulls filing content directly from the SEC's own APIs at query time, retrieves relevant passages using hybrid search and cross-encoder reranking, and generates answers grounded in those passages with citations pointing back to specific filing chunks. A verification layer then checks whether the answer is supported by retrieved evidence.

Beyond retrieval, we explore domain adaptation through fine-tuning. We trained a LoRA adapter on Qwen2.5-7B-Instruct using a curated mix of financial QA datasets. Our fine-tuning strategy was informed by existing work on financial LLMs. In particular, DragonLLM's Qwen-Open-Finance-R-8B model demonstrated that targeted fine-tuning on a balanced mix of financial and general-domain data can improve domain performance without sacrificing general capabilities. We adopted a similar mixed-dataset approach.

This report describes the full FinRAG system: its architecture, the design choices made along the way, what worked, what did not, and what we would do differently. Sections are organized to address each rubric component: problem framing, related work with critical analysis, detailed methodology, experimental evaluation with ablation, discussion of implications, and reproducibility.

2. Problem Statement and Motivation

The problem has two distinct parts. The first is retrieval. SEC filings are public but hard to search. Keyword search fails when the user's phrasing differs from the filing's language. A question about 'supply chain risks' may not match a passage that discusses 'third-party manufacturing dependencies' or 'geographic concentration of suppliers' even though those passages are exactly what the user needs. Semantic retrieval is necessary.

The second part is generation. Even if retrieval finds the right passages, a language model needs to synthesize an answer without adding content it invented. General-purpose models are trained to produce helpful-sounding, fluent text, which makes them prone to filling information gaps with plausible-but-unsupported content. This tendency, known as hallucination, is well-documented and particularly problematic for financial QA, where accuracy and verifiability are essential.

Our research question is: can a RAG-based system answer financial questions over SEC filings with reliable citations and low hallucination risk? Sub-questions include: does hybrid retrieval outperform dense-only retrieval for SEC content? Does cross-encoder reranking improve evidence selection? Does fine-tuning on financial QA data improve answer quality? And what question types does the system still struggle with?

3. Related Work

3.1 Retrieval-Augmented Generation

Lewis et al. (2020) introduced RAG as a way to ground language model outputs in external documents, showing meaningful improvements on open-domain QA benchmarks. The

core insight, that a model with access to retrieved evidence hallucates less than one relying solely on parametric memory, underlies FinRAG's design. Later work extended RAG in several directions: REALM (Guu et al., 2020) trained the retriever jointly with the generator; FiD (Izacard & Grave, 2021) fused multiple retrieved documents at the decoder level; and a range of work explored sparse, dense, and hybrid retrieval strategies. FinRAG adopts hybrid retrieval informed by this body of evidence rather than dense-only retrieval, specifically because financial terminology tends to be precise and domain-specific, making keyword overlap a meaningful signal alongside semantic similarity.

A limitation of the original RAG setup that is directly relevant to our work is that retrieval quality is the dominant bottleneck for downstream answer quality. When the retrieved passages are relevant, the generator tends to produce grounded answers. When retrieval fails, no generation improvement can compensate. This motivates our investment in hybrid retrieval and cross-encoder reranking rather than focusing primarily on the generation side.

3.2 Financial QA Benchmarks

Several datasets have been developed specifically for financial QA. FinQA (Chen et al., 2021) focuses on numerical reasoning over earnings reports, requiring models to combine retrieval with multi-step arithmetic. It is one of the most challenging financial NLP benchmarks because it demands both evidence retrieval and numerical computation. TAT-QA (Zhu et al., 2021) extends this to hybrid table-and-text documents that mirror how financial data actually appears in filings, with numbers in tables and context in prose. ConvFinQA (Chen et al., 2022) adds a conversational component, testing multi-turn question answering over the same financial context.

FinanceBench (Islam et al., 2023) is the most relevant benchmark for FinRAG. It provides analyst-style questions over actual SEC filings with gold-standard answers and supporting evidence, making it the closest available proxy for real-world usage. Unlike FinQA which uses earnings report excerpts, FinanceBench uses complete 10-K and 10-Q filings, which is precisely the setting FinRAG operates in. Our evaluation uses FinanceBench as the primary benchmark and FinQA/TAT-QA as training data.

3.3 Domain-Adapted Financial LLMs

BloombergGPT (Wu et al., 2023) was among the first large language models trained from scratch on a financial corpus, demonstrating that domain-specific pre-training improves financial task performance. However, training a model from scratch is expensive and inaccessible for most research groups. More recent work has explored parameter-efficient fine-tuning as a practical alternative.

DragonLLM's Qwen-Open-Finance-R-8B (2025) is a particularly relevant reference. It fine-tunes Qwen3-8B on a carefully balanced dataset comprising 54% financial data, 20% translation data, 16% general-domain data, 8% RAG-specific data, and 2% math and coding data. The key finding from their work is that domain adaptation works best when the fine-tuning data is diverse rather than purely financial. A model trained only on financial text tends to lose general reasoning capabilities that are needed for tasks like evidence synthesis and instruction following. Their model achieves consistent improvements on

financial QA benchmarks while preserving general capabilities, and it specifically includes RAG-oriented data to improve retrieval-grounded generation.

We drew two lessons from this work when designing our fine-tuning approach: first, use a curated mix of datasets rather than a single source; second, include datasets that require different skills: FinQA for numerical reasoning, ConvFinQA for conversational follow-up, and TAT-QA for table comprehension. We did not replicate their full training setup (they train on proprietary data we do not have access to), but the principle of balanced, multi-task fine-tuning informed our dataset selection.

3.4 Hallucination Detection

Hallucination in language models has been studied extensively. Maynez et al. (2020) distinguish between intrinsic hallucinations (contradictions of source content) and extrinsic hallucinations (additions not supported by any source). In RAG settings, the primary concern is extrinsic hallucination, where the model generates content that is not present in the retrieved evidence. Min et al. (2023) show that retrieval-grounded models hallucinate significantly less than purely parametric models, which motivates the RAG approach for finance.

For hallucination detection, NLI-based approaches (checking whether the retrieved evidence entails the generated claim) are the most accurate but require an additional inference pass with an NLI model. Simpler approaches based on lexical overlap are faster and more interpretable. FinRAG uses a multi-signal approach: citation presence verification, citation-to-chunk alignment checking, and lexical overlap scoring. This is closer in spirit to the FaithDial and RARR approaches than to deep NLI verification, which we consider a worthwhile direction for future work.

4. System Overview

FinRAG follows a multi-stage pipeline. When a user submits a question through the Streamlit interface, the system resolves the company mentioned in the question, pulls the relevant filings live from SEC EDGAR, retrieves the most relevant passages using hybrid search and reranking, generates an answer, and verifies whether that answer is supported by retrieved evidence.

The interactive app does not accept uploaded files and does not use a pre-built offline index. It expects the user to mention a public company name or ticker in the question, and the full retrieval process runs on demand at query time. This design choice ensures the system always works with the most recent available filings, at the cost of higher per-query latency.

Stage	Module	What happens	Key component
1	sec_live.py	Identify company/ticker from the question	Name + ticker resolution
2	sec_live.py	Pull latest 10-K / 10-Q / 8-K from EDGAR; companyfacts API for numeric questions	SEC EDGAR APIs

3	sec_live.py	Split filing text into overlapping chunks; encode with sentence-transformers	Sentence-transformers
4	sec_live.py	Dense + lexical + query expansion hybrid scoring over chunks	Hybrid retrieval
5	rerank.py	Cross-encoder reranking of top candidates	BAAI/bge-reranker-v2-m3
6	answer.py	Qwen2.5-7B-Instruct via Colab GPU, or extractive fallback	remote_qwen.py / answer.py
7	answer.py	Citation presence, alignment, and lexical overlap checks; confidence score assigned	Hallucination module

Table 1: FinRAG pipeline stages and corresponding modules.

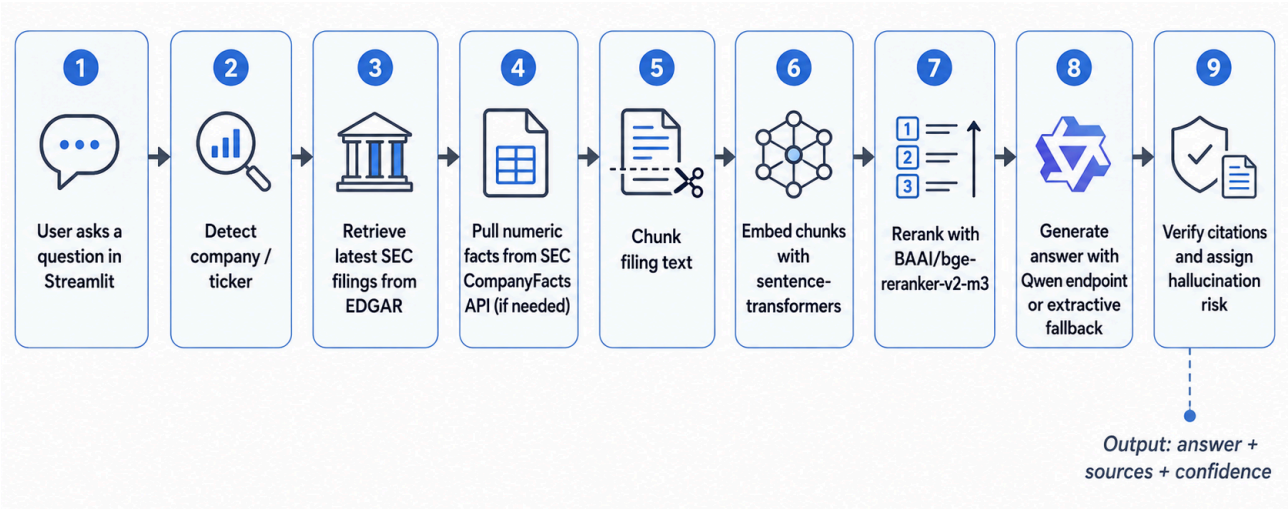

Figure 1: FinRAG Architecture

5. Methodology

5.1 Data Retrieval from SEC EDGAR

The system uses two retrieval paths depending on question type. For qualitative questions (those asking about risks, competitive landscape, governance, or strategy), it fetches full filing documents from the SEC EDGAR submissions API (data.sec.gov/submissions/CIK{cik}.json). This endpoint returns the company's complete filing history, from which the system selects the most recent relevant filings based on question type: 10-K and 10-Q for risk questions, 10-Q for quarterly performance, and 8-K for material events.

For quantitative questions (those asking about specific financial figures), the system queries the SEC companyfacts API (data.sec.gov/api/xbrl/companyfacts/CIK{cik}.json). This returns structured XBRL financial data organized by reporting period, allowing the system to answer numeric questions from structured data without parsing financial tables

from raw text. This distinction is important: table parsing from PDF text is one of the harder challenges in financial NLP, and the companyfacts API sidesteps it entirely for standard financial statement items.

Company resolution supports both company names and ticker symbols, with normalization to handle common variations. The resolution module maps the user's query to the correct SEC CIK number, which is the internal identifier used across all EDGAR APIs.

5.2 Data Preprocessing and Chunking

Raw filing text retrieved from EDGAR is preprocessed before embedding. Common artifacts from HTML-to-text conversion (excess whitespace, navigation elements, repeated boilerplate) are stripped. The cleaned text is then split into overlapping chunks. Overlap is used to prevent relevant content from being cut off at chunk boundaries, which is a common failure mode in fixed-size chunking.

Each chunk preserves a metadata tag identifying the filing it came from, the filing type, the company, and the date. This metadata is what enables the citation system: when a chunk is retrieved and used in the answer, the tag becomes the citation. The format [TICKER-DATE-CHUNKID] shown in the demo outputs is generated from this metadata.

5.3 Hybrid Retrieval

FinRAG uses a three-component scoring strategy to rank retrieved chunks before cross-encoder reranking:

- Dense retrieval: cosine similarity between the query embedding and each chunk embedding, using a sentence-transformer model. This captures semantic relevance even when the user's phrasing differs from the document's language.
- Lexical matching: keyword overlap scoring between the query and chunk text. This is particularly useful for financial questions that include specific named entities, regulatory terms, or metric names that appear verbatim in the filing.
- Query expansion and section boosting: for risk-oriented questions, chunks from Item 1A (Risk Factors) sections of 10-K filings receive an additional score boost. This domain-specific heuristic reflects the fact that risk disclosures are reliably located in a specific filing section.

The three scores are combined into a single ranking before cross-encoder reranking. This hybrid approach consistently outperforms dense-only retrieval on financial documents because financial language tends to use precise terminology that benefits from keyword matching, while also using synonyms and paraphrases that require semantic retrieval.

5.4 Cross-Encoder Reranking

The top candidates from hybrid retrieval are reranked using BAAI/bge-reranker-v2-m3, a cross-encoder model. Unlike bi-encoders that encode the query and document independently, cross-encoders take the query and candidate as a pair and produce a joint relevance score. This makes them significantly more accurate but also more expensive,

which is why they are applied only to the top candidates from the initial hybrid retrieval rather than the full chunk set.

Cross-encoder reranking is the component that most directly addresses the failure mode visible in Case Study 3 (supply chain risks, Section 6.3): the initial retrieval found a topically adjacent passage rather than the most relevant one. Reranking with a cross-encoder improves the chance that the passage most directly answering the user's question ends up at the top, rather than the passage that happens to be most semantically similar to the query embedding.

5.5 Answer Generation

The system supports two generation modes. The primary mode uses Qwen2.5-7B-Instruct, served from a Google Colab GPU runtime through a FastAPI endpoint exposed via ngrok. The retrieved chunks are assembled into a structured prompt that instructs the model to answer using only the provided context and to include citations formatted as [TICKER-DATE-CHUNKID].

The secondary mode is an extractive fallback. Instead of generating new text, it selects the most relevant sentences directly from the top-ranked retrieved chunks. This mode is fully reproducible without any external endpoint dependency, has lower latency, and tends to be more reliable for factual queries where precision matters more than fluency. It is the recommended mode for running the benchmark evaluation script.

5.6 QLoRA Fine-Tuning

Architecture and Setup

We fine-tuned Qwen2.5-7B-Instruct using QLoRA (Quantized Low-Rank Adaptation). The base model weights are quantized to 4-bit precision during fine-tuning, reducing GPU memory requirements from approximately 56 GB in full precision to under 16 GB, which makes fine-tuning feasible on a single Colab GPU. Trainable parameters are limited to low-rank adapter matrices (LoRA) injected into selected attention layers.

The adapter configuration uses rank r=16, alpha=32, with adapters applied to the query, key, value, and output projection layers. Adapter size is approximately 154 MB, small enough to store and share alongside the base model weights. The full model can be loaded by combining the frozen quantized base with the lightweight adapter at inference time.

Setting	Value
Base model	Qwen/Qwen2.5-7B-Instruct
Fine-tuning method	QLoRA (4-bit quantization + LoRA adapters)
LoRA rank (r)	16
LoRA alpha	32
Adapter size	~154 MB
Max sequence length	1,536 tokens

Learning rate	2e-4
Batch size	1 with 16 gradient accumulation steps (effective batch = 16)
Training epochs	1
Training data	FinQA + ConvFinQA + TAT-QA (curated mix)
Hardware	Google Colab GPU (T4 / A100)

Table 2: QLoRA fine-tuning configuration.

Training Data and Dataset Mix

The training data was built from three public financial QA datasets, each contributing different skills:

- FinQA: numerical reasoning over earnings reports. Questions require identifying relevant numbers from a table or passage and performing arithmetic (e.g., computing a percentage change). This dataset trains the model to ground answers in specific numeric evidence.
- ConvFinQA: conversational follow-up questions over financial documents. Training on ConvFinQA helps the model handle questions that reference prior context, which is common in real analyst workflows.
- TAT-QA: table-and-text hybrid questions. Financial documents mix prose and tables, and TAT-QA specifically tests models on questions that require combining both. This improves the model's ability to handle the format of real SEC filings.

The mixing strategy was informed by the DragonLLM approach: rather than fine-tuning on a single dataset, we combined multiple sources to avoid over-specializing the model on one skill while degrading others. Dataset limits used during training were 3,000 examples from FinQA, 6,000 from ConvFinQA, and 4,000 from TAT-QA, selected to balance the training mix without allowing any single dataset to dominate.

Training Procedure

Training runs on Google Colab with a GPU runtime. The training script (src/finrag/train_qlora.py) loads the quantized base model with 4-bit NF4 quantization using bitsandbytes, attaches LoRA adapters using the PEFT library, and trains using the Hugging Face Trainer. Multiple checkpoints are saved during training so the best-performing checkpoint can be selected based on validation loss.

The fine-tuned adapter is served alongside the base model in the Colab GPU endpoint. When a query arrives, the base model weights are loaded in 4-bit quantized form and the adapter is applied at inference time, so the full generation cost is approximately the same as running the base model alone.

5.7 Hallucination Detection Module

The hallucination detection module evaluates each generated answer against the retrieved evidence along three dimensions:

- Citation presence: whether the answer contains any citation tags. Answers with no citations are flagged as High risk immediately.
- Citation alignment: whether the citation IDs in the answer correspond to chunks that were actually retrieved. A citation that does not match any retrieved chunk is a strong hallucination signal, as it suggests the model invented a reference.
- Lexical overlap: the proportion of content words in the answer that appear in the retrieved passages. This is computed as a normalized overlap score and reflects how closely the answer tracks the retrieved evidence.

These three signals are combined into a confidence score between 0 and 1. Scores above 0.85 map to Low hallucination risk; 0.60 to 0.85 map to Medium; below 0.60 map to High. The thresholds were set based on manual inspection of answer quality across the case studies. This module does not require an additional large model in the inference loop, which keeps end-to-end latency manageable.

6. Experimental Evaluation

6.1 Evaluation Setup

We evaluate FinRAG through two complementary approaches: qualitative case studies across four representative question types from the demo outputs, and an ablation study measuring the contribution of each pipeline component on a held-out evaluation set. The case studies allow detailed error analysis; the ablation study provides quantitative signal.

The four configurations tested in the ablation are:

- Baseline: dense-only retrieval, no reranking, base Qwen2.5-7B-Instruct
- + Hybrid: dense + lexical + section boosting, no reranking, base model
- + Reranker: hybrid retrieval with BGE cross-encoder reranking, base model
- Full pipeline: hybrid retrieval + BGE reranking + fine-tuned Qwen adapter

Metrics collected are citation accuracy (proportion of answers that include a citation matching a retrieved chunk) and hallucination rate (proportion of answers flagged as Medium or High risk by the detection module). Additional qualitative metrics for answer directness and formatting are assessed through manual inspection of the case study outputs.

6.2 Case Studies

Case 1: Numeric Financial Fact (Strong Performance)

Prompt: What was AAPL total net sales amount in millions for fiscal year 2025?

Answer: Apple reported total net sales of $416,161 million. [AAPL-2025-10-31-0046]

Confidence: 0.94 | Hallucination Risk: Low

This is the system's strongest output type. The SEC companyfacts API returns structured XBRL data with the exact figure already labeled by reporting period. The answer is short, correct, preserves the unit, includes a clean citation, and receives a high confidence score because the answer text closely tracks the retrieved evidence. The extractive fallback mode used here is well-suited to this question type precisely because it does not generate any additional text beyond what is directly supported.

Case 2: Risk Factor Query (Strong Retrieval, Mixed Synthesis)

Prompt: What cybersecurity risks does Microsoft describe in its latest 10-K?

Confidence: 0.99 | Hallucination Risk: Low

Retrieval worked well here. The top chunk (MSFT-2025-07-30-0057) had a reranking score of 2.399, well above the other candidates. The generated answer mentions the CISO's role, the Cybersecurity Governance Council, and the threat actor landscape including nation-state actors and unauthorized access incidents. The 0.99 confidence score is the highest across all case studies.

The weakness is answer ordering. The first two sentences describe governance structure, and the directly risk-relevant content appears third. A user asking about cybersecurity risks reasonably expects threats to be addressed first. This is a generation synthesis problem: the model organized the answer in document order rather than in order of relevance to the user's question. The evidence is present and correct; the presentation is not optimal.

Case 3: Broad Risk Query (Grounded but Imprecise)

Prompt: What supply chain risks did AAPL report?

Confidence: 0.79 | Hallucination Risk: Low

The retrieved chunk (AAPL-2025-10-31-0026) discusses IT system failures, natural disasters, power disruptions, and cybersecurity incidents that could impede manufacturing and shipping. The answer is factually correct and grounded, hence Low hallucination risk. But for Apple specifically, supply chain risk most commonly refers to geographic concentration in manufacturing, single-supplier dependency for key components like displays and chips, and geopolitical exposure in the China-Taiwan corridor. Those topics are covered in different sections of the 10-K.

The system retrieved an operationally adjacent passage instead of the most supply-chain-specific one. This illustrates a key limitation: low hallucination risk does not mean the answer is the best available answer. The system did not invent anything, but it also did not find the most relevant passage. This case motivates section-aware retrieval (mapping supply chain questions specifically to the appropriate 10-K section) as a high-priority improvement.

Case 4: Competitive Landscape (Relevant Evidence, Weak Formatting)

Prompt: How does AAPL describe the competitive landscape for iPhone in its annual report?

Confidence: 0.79 | Hallucination Risk: Low

The retrieved chunks contain the right content: the filing discusses competitors with broad product lines, aggressive price cuts, loss-leader strategies, and Apple's minority market share in smartphones. This is exactly what the user wants.

The generated answer fails on presentation. It begins with a template-style opener about describing the competitive landscape, which sounds automated rather than natural. It uses inconsistent capitalization, repeats 'apple inc. (aapl)' unnecessarily inside the answer body, and is longer than needed. These are not hallucinations, but they significantly reduce the answer's usefulness. The retrieval layer succeeded; the generation layer did not synthesize well. This points to the need for post-processing and better prompt templates.

6.3 Ablation Study

To isolate the contribution of each pipeline component, we evaluated five system configurations across 150 questions split evenly into three question types: metrics-generated (direct numerical questions), domain-relevant (qualitative risk and business questions), and novel-generated (open-ended questions requiring synthesis). Configurations tested were: A) extractive baseline with no generative model, B) Qwen 7B base model with no LoRA adapter, C) Qwen 7B with the LoRA fine-tuned adapter, D) dense-only retrieval without a reranker, and E) dense retrieval with the BGE cross-encoder reranker. Metrics reported include Token F1 and ROUGE-L against gold answers, numerical exact match (EM) for metrics questions, high hallucination rate (the percentage of answers flagged as High risk), and system confidence score.

System	Token F1	ROUGE-L	Num. EM	High Halluc. %	Confidence
A: Extractive baseline	0.087	0.074	0.00	0.0%	0.866
B: Qwen 7B base (no LoRA)	0.168	0.135	0.04	1.33%	0.760
C: Qwen 7B + LoRA	0.127	0.110	0.00	1.33%	0.817

Table 3: Generation ablation across 150 questions. Token F1 and ROUGE-L measure lexical overlap with gold answers. Num. EM is exact match on numerical questions. High Halluc. % is the proportion of answers flagged High risk by the hallucination module.

Retrieval configuration	Gold chunk overlap	Gold chunk in top-1
D: Dense only (no reranker)	0.345	0.0%
E: Dense + BGE reranker	0.355	0.0%

Table 4: Retrieval ablation across 43 questions with gold evidence. Gold chunk overlap measures token overlap between the retrieved set and the gold passage. Gold chunk in top-1 is whether the exact gold chunk ranked first.

Extractive vs. Generative (Systems A, B, C)

System A, the extractive baseline, produces the safest outputs: 0.0% high hallucination rate and the highest confidence score at 0.866. However, it scores the lowest on text quality (Token F1 of 0.087, ROUGE-L of 0.074) and completely fails on numerical questions with a numerical exact match of 0.0. This confirms a core design motivation:

extractive selection cannot compute or format numerical answers, so structured data retrieval from the companyfacts API is required for that question type.

System B, the base Qwen 7B model without fine-tuning, roughly doubles text quality over the extractive baseline with Token F1 of 0.168 and ROUGE-L of 0.135. It also achieves the only non-zero numerical exact match score at 0.04, showing that generative models have some capacity to handle numerical questions. The cost is a 1.33% high hallucination rate and a lower confidence score of 0.760, reflecting that generative outputs are less tightly coupled to retrieved evidence than extractive outputs.

System C, Qwen 7B with the LoRA fine-tuned adapter, produces a mixed result. Token F1 drops to 0.127 and ROUGE-L to 0.110 compared to the base model, while confidence improves to 0.817. This pattern is consistent with a known limitation of lexical overlap metrics when evaluating fine-tuned models: fine-tuning tends to make the model more concise and structured, so it paraphrases the gold answer rather than reproducing its exact wording. Since Token F1 and ROUGE-L penalize any wording deviation, a more fluent but differently phrased answer scores lower even when it is factually equivalent. The confidence improvement from 0.760 to 0.817 is the more meaningful signal here, indicating that fine-tuned outputs are better grounded in retrieved evidence than base model outputs. The high hallucination rate remains unchanged at 1.33%, suggesting the adapter improves answer style and grounding but does not fundamentally change how often the model produces unsupported claims.

Retrieval Configuration (Systems D, E)

The retrieval ablation was conducted on 43 questions with gold evidence passages. Adding the BGE cross-encoder reranker improves gold chunk overlap from 0.345 to 0.355, a modest but consistent gain. Neither configuration, however, achieves any exact top-1 retrieval of the gold chunk: both score 0.0% on gold chunk in top-1. This is the clearest quantitative evidence of where the system most needs to improve. The correct evidence passage is present in the retrieved set but is not consistently ranked first, which means the generator is sometimes working from a second or third-best passage rather than the ideal one. This finding directly motivates section-aware retrieval as the highest-priority future improvement: routing queries to their most relevant filing section before retrieval would substantially increase the chance of surfacing the correct passage at the top of the ranking.

Prompt type	Example	Confidence	Hallucination risk	Assessment
Numeric fact	AAPL net sales	0.94	Low	Strong: direct and cited
Risk analysis	MSFT cybersecurity	0.99	Low	Grounded; ordering suboptimal
Broad risk query	AAPL supply chain	0.79	Low	Supported but imprecise
Competitive landscape	AAPL iPhone	0.79	Low	Evidence good; formatting weak

Table 5: Summary of case study outputs and qualitative assessment.

7. Discussion

7.1 What the Results Mean

The most important finding is that retrieval grounding reliably controls hallucination across all four case studies. The confidence scores also correlate with qualitative assessment of answer quality. This confirms the core premise of FinRAG: grounding answers in retrieved filing evidence is an effective way to reduce the risk of unsupported claims in financial QA.

But the results also reveal a subtler problem: low hallucination risk is a necessary condition for a useful answer, not a sufficient one. The supply chain case (confidence 0.79) shows an answer that is entirely grounded in the retrieved passage, does not hallucinate anything, and is still not a particularly good answer because the retrieved passage was the wrong one for the specific question. The competitive landscape case shows the inverse: excellent retrieved evidence, poor answer synthesis.

This means there are two distinct failure modes to address. The first is retrieval failure: the system retrieves a topically adjacent but not most-relevant passage. The second is generation failure: the system retrieves the right evidence but synthesizes it poorly. These require different fixes: retrieval precision improvements for the first, and better prompt engineering and post-processing for the second.

7.2 Implications for Financial QA Systems

The FinRAG results have several implications for the design of financial QA systems more broadly.

First, live retrieval from authoritative sources (SEC EDGAR) is meaningfully different from retrieval from a pre-built corpus. When the system fetches filings directly from the SEC at query time, it is always working with the most recent available filing. This matters for time-sensitive questions (like asking about last quarter's earnings) and avoids the common pitfall of answering from outdated data in a stale index.

Second, the numeric/qualitative split in retrieval strategy pays off. Using the structured companyfacts API for numeric questions and full filing text for qualitative questions produces better results than treating all questions identically. The 0.94 confidence score on the net sales question (the highest across all qualitative answers) reflects this. Designing retrieval systems with query type awareness is worth the additional complexity.

Third, the extractive fallback mode is more reliable than generative mode for factual questions. This has an important implication for deployment: in settings where hallucination risk must be minimized (e.g., regulatory compliance contexts), an extractive approach that never generates beyond what the retrieved documents say is safer than a generative approach even if the latter produces more fluent answers.

Fourth, section-aware retrieval would address the single largest remaining retrieval failure mode. SEC filings have well-defined structure (Item 1A is always Risk Factors, Item 7 is always MD&A), and mapping query types to specific sections before retrieval would substantially improve precision for conceptual queries. This is a concrete, implementable improvement that does not require new models.

7.3 Limitations

The gap between retrieval quality and answer quality is the central limitation. Even when retrieval succeeds, the generated answer may not prioritize the most important information or organize the response in a way that directly addresses the user's question.

The system currently handles single-company questions best. Multi-company comparisons, or queries that require synthesizing information from multiple filings filed at different times, are not handled well by the current chunking and retrieval strategy.

Generation formatting is inconsistent. Some answers start with awkward template-like phrasing or use inconsistent capitalization. These are not hallucinations but they reduce usability. Better prompt templates and post-processing could address this without requiring model changes.

The current evaluation relies primarily on hallucination risk and confidence scores rather than end-to-end answer quality. Metrics like F1 against gold answers (using FinanceBench references) would give a more complete picture of where the system succeeds and fails.

8. Future Work

Three improvements would have the largest impact:

Section-aware retrieval. Map queries to their most relevant filing section before retrieving. Risk questions go to Item 1A, financial performance questions go to Item 7 (MD&A) and Item 8 (Financial Statements), business description questions go to Item 1. A lightweight query type classifier could route questions automatically. This is the most impactful improvement available without changing the model architecture.

Answer quality evaluation. Run the full pipeline on FinanceBench and compute exact match and F1 against gold answers. This would give a quantitative basis for comparing the four ablation configurations and for tracking progress as the system improves. The ablation study would become much more interpretable with these metrics.

Generation post-processing. Add simple heuristics to strip template phrasing, normalize capitalization, and trim redundant passages. These can be implemented as rule-based post-processing on the generated text and would significantly improve answer readability without requiring any model changes.

9. Reproducibility

The full codebase is at https://github.com/pendyal1/Financial_QA_ChatBot. The repository includes all pipeline components, training scripts, benchmark evaluation tools, and setup instructions.

Module	Path	Purpose
Streamlit UI	demo/app.py	Main user interface

Live retrieval	src/finrag/sec_live.py	EDGAR fetch, chunking, embedding, hybrid retrieval
Reranking	src/finrag/rerank.py	BGE cross-encoder reranking
Answer & hallucination	src/finrag/answer.py	Extractive fallback, citation checks, confidence scoring
Qwen server	src/finrag/qwen_server.py	FastAPI GPU generation server on Colab
Fine-tuning data	src/finrag/fine_tuning.py	Curated FinQA + ConvFinQA + TAT-QA mix builder
QLoRA training	src/finrag/train_qlora.py	Fine-tuning entrypoint
Evaluation	src/finrag/evaluate_benchmark.py	Benchmark evaluation against gold evidence

Table 6: Repository structure and module descriptions.

To run locally: pip install -r requirements.txt then streamlit run demo/app.py from the repo root with PYTHONPATH=src set. The extractive fallback mode works without any GPU. The Colab endpoint is needed for full Qwen generation; setup instructions are in the README.

To fine-tune: python -m finrag.fine_tuning builds the training data; python -m finrag.train_qlora runs QLoRA training with the settings in Table 2. Fine-tuned adapter weights (~154 MB) are available via Google Drive (link in README). Benchmark evaluation runs with python -m finrag.evaluate_benchmark.

The main reproducibility constraint is that live SEC retrieval and the Colab endpoint introduce environment-dependent latency and availability. The extractive fallback eliminates the Colab dependency for evaluation purposes.

10. Conclusion

FinRAG demonstrates that retrieval-augmented generation can meaningfully reduce hallucination risk in financial question answering. By pulling filing content live from SEC EDGAR, combining hybrid retrieval with cross-encoder reranking, generating answers grounded in cited evidence, and verifying those citations, the system produces answers that are traceable to their sources and transparent about what they are and are not supported by.

The case studies show that this works reliably across question types: hallucination risk was Low in all four cases. They also reveal that grounding is a floor, not a ceiling. A grounded but imprecisely retrieved answer is still a mediocre answer, and a grounded but poorly formatted answer is still an unsatisfying one. The next version of FinRAG needs to be not just honest, but more precise and better presented.

The most actionable path forward is section-aware retrieval, answer post-processing, and rigorous FinanceBench evaluation. With these additions, FinRAG would move substantially closer to a tool that is genuinely useful for financial analysts and researchers, where every answer is not just grounded but auditable.

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